Embryology of the Cardiovascular System

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Introduction & timing

The **CVS (heart and blood vessels) is the earliest functional system in the body**, developing from **18 days to 12 weeks**.

Why so early? The tiny early embryo can meet its needs by **simple diffusion** of nutrients and oxygen. But around the **3rd week**, it grows **beyond the diffusion threshold** – the inner cells can no longer be reached fast enough. A **delivery system is needed**, so the CVS forms first, starting at **day 18**, and is largely complete by **12 weeks**.

The five sources of the CVS

Source	Contribution
Splanchnic mesoderm	Primordium (foundation) of the heart
Paraxial & lateral mesoderm	Blood vessel walls, connective tissue
Otic placodes	Minor (relates to ear; listed for completeness)
Neural crest cells	Outflow tract septation, great vessel walls
Primordium of the labyrinth of the internal ear	Minor/historical

> **The two to remember: - Splanchnic mesoderm -> heart muscle and lining - Neural crest cells -> the outflow tract** (defects here cause **TGA, TOF, Truncus arteriosus**)

Neural crest cells recur throughout this lecture – they are responsible for some of the most important and most exam-tested congenital heart defects.

The cardiogenic field

By the **3rd week**, cardiac progenitor cells called **myoblasts** cluster in the **cardiogenic field** (the area where the heart will form).

- **Myoblasts** – future heart-muscle cells; they organize into cord-like structures called **angioblastic cords**
- **Angioblastic cords** – hollow out and fuse to form the **primitive heart tube**
- **Position** – the cardiogenic field sits **anterior (in front) of the endodermal primitive pharynx**
- **Cephalad origin** – the heart starts forming **up near the head (cephalad)**, then **migrates downward** to the chest as the embryo folds craniocaudally in weeks 3-4

Formation of the heart tube

By the **4th week**, the two angioblastic cords have hollowed and **fused into one straight tube – the primitive heart tube**. Before it can be shaped, it must be **anchored** at both ends:

Anchor point	Structure
Caudal end (bottom)	**Septum transversum** (a block of mesoderm that later helps form the diaphragm)
Cephalad end (top)	**The (6) arterial arches**

Dilatation of the tube becomes visible on **days 21-24**.

The tube is already polarized

- **Venous end = caudal (bottom)** -> will become the **atria**
- **Arterial end = cephalad (top)** -> will become the **ventricles and outflow tract**

  This polarity seems backwards – veins at the bottom, arteries at the top – but the embryo is about to **loop and fold**, rearranging everything into the adult arrangement.

Looping

The problem: a straight tube anchored at both ends inside a limited space (the pericardial cavity) is **growing rapidly but has nowhere to elongate**.

The solution: it **bends and loops on itself**, like a garden hose coiling when you push both ends together. This looping is **genetically programmed** and happens in a **specific direction** – the wrong direction gives **dextrocardia**.

Stage 1 – Atrial looping

- **Site:** between the primitive atrium (PA) and the interventricular region (IV)
- **Angle:** a right-angle bend
- **Mechanism – differential growth:** the posterior-left side grows slowly, the anterior-right side grows fast; this unequal growth forces the bend

Type	Result
D-loop (dextro)	Right ventricle ends up front and to the right – **NORMAL**
L-loop (laevo)	Left ventricle ends up front and to the left – **ABNORMAL**

(Note: the lecturer's slide labels these in a way that can confuse – the key point is that one direction is normal and the opposite produces a malpositioned heart / dextrocardia.)

Stage 2 – Ventricular looping

- **Site:** between the interventricular region (IV) and the outlet ventricle (OV)
- **Angle:** a dramatic **180° flexure**
- **Result:** the tube now resembles the definitive heart – you can identify where the chambers will be

> **What looping achieves:** the atria (which started at the bottom/venous end) are repositioned **behind and above the ventricles**; the ventricles become **anterior and inferior**. This is the adult arrangement – **atria on top/behind, ventricles below/in front**.

The primitive heart tube – sinus venosus, primitive atrium and ventricle, bulbus cordis, truncus arteriosus

Looping of the heart tube

Convergence and wedging

After looping, the heart looks roughly right – but the **inlets** (blood entering the atria) and **outlets** (blood leaving the ventricles) are **not yet aligned over their correct ventricles**. Two repositioning movements fix this before septation.

Convergence – the **AV canal (inlet)** and the **bulbus cordis (outlet)** move **closer together**, into the same plane, so they can connect to the right chambers (this happens while the AV canal is already septating).

Wedging – a two-part movement:

Structure	Direction	Result
AV canal	shifts to the **right**	each AV orifice (mitral, tricuspid) sits over its own ventricle
Bulbus cordis	shifts to the **left**	half of the outlet sits over each ventricular outflow tract

After wedging: **tricuspid over RV , mitral over LV , aorta wedged between the AV valves over the LV , pulmonary trunk over the RV .**

> **When convergence/wedging fail: - Double Outlet Right Ventricle (DORV)** – both aorta and pulmonary trunk over the RV (wedging failed) **- Double Inlet Left Ventricle (DILV)** – both AV valves drain into the LV (convergence abnormal)

Veins associated with the heart

Three paired sets of veins drain into the **sinus venosus** (the most caudal part of the heart tube):

Vein	Function	Adult fate
Vitelline vein	Returns deoxygenated blood from the **yolk sac**	**Hepatic veins and portal vein**
Umbilical vein	Carries **oxygenated** blood from the **placenta**	**Ductus venosus -> ligamentum venosum** at birth
Common cardinal vein	Returns deoxygenated blood from the body	Forms the **SVC and IVC** (right-sided dominance)

> **Umbilical vein carries oxygenated blood** – opposite of what a "vein" usually does – because in fetal life **the placenta is the lung**. The **ductus venosus** lets that oxygenated

> blood **bypass the liver** and go straight to the heart; at birth it closes to become the **ligamentum venosum**.

The cardinal system – right-sided dominance

The cardinal system is paired (left and right), but the sides have **unequal fates**: the **right side largely persists** to form the major systemic veins; the **left side mostly regresses**. This is why adults have a **single SVC and IVC, both on the right**.

Each common cardinal vein = **anterior cardinal** (drains the head) + **posterior cardinal** (drains the lower body).

Vessel	Adult structure
Right anterior cardinal vein	**Superior vena cava (SVC)**
Left anterior cardinal vein	Largely regresses

The **IVC** has a complex origin – assembled from **four segments**: hepatic veins, right subcardinal vein, right supracardinal vein, and the subcardinal-supracardinal anastomosis. Failure of any segment gives a specific anomaly (**absent infrarenal IVC, interrupted IVC with azygos continuation, double IVC**).

> **Everything is right-sided** – SVC from the right anterior cardinal, IVC from right-sided vessels; the left equivalents disappear. This **right-sided dominance** is a consistent theme.

The cardinal, vitelline and umbilical veins draining into the sinus venosus

Atrial septation

The atria **cannot be completely sealed** in fetal life – the fetus doesn't use its lungs, so oxygenated placental blood needs to **bypass the pulmonary circulation** and reach the left heart directly. Nature's solution: **divide the atria but leave a controlled hole – the foramen ovale** – that closes only at birth.

Stages of atrial septation and the resulting septal defects

- **Septum primum** – grows **down from the roof** of the primitive atrium toward the **endocardial cushions**
- **Foramen primum** – the gap between the free edge of septum primum and the endocardial cushions (**first hole**; lets blood pass right -> left while the septum grows)
- **Foramen secundum** – as septum primum reaches the floor and closes the foramen primum, **perforations appear higher up in septum primum** and coalesce into the **second hole** (so flow is maintained)
- **Septum secundum** – an **invagination of the roof**, growing down to the **right of septum primum**; it is **thicker and more muscular** (to withstand the high right-atrial pressure of fetal life)
- **Foramen ovale** – septum secundum grows down but **deliberately stops short**, leaving this gap

The foramen ovale – septum primum acts as a one-way flap valve

The foramen ovale as a one-way valve

In fetal life: right atrial pressure > left (lungs not working) -> pressure pushes septum primum **leftward**, opening the foramen ovale -> blood flows **right -> left**, bypassing the lungs

At birth: first breath -> lungs expand -> pulmonary pressure drops -> **left atrial pressure rises above right** -> septum primum is pushed **rightward against septum secundum**, covering the foramen ovale -> the two septa **fuse** -> foramen ovale closes -> becomes the **fossa ovalis**.

Septum primum and septum secundum – RA, LA, RV, LV

The endocardial cushions

Masses of tissue in the AV canal region that: help close the foramen primum; contribute to the **mitral and tricuspid valves**; and contribute to the **membranous part of the interventricular septum**.

> **Failure of endocardial cushion development -> AV septal defects (AVSD)** – common in **Down syndrome**.

When atrial septation fails

Defect	Type	Cause
ASD secundum	Most common ASD	Excessive resorption of septum primum / foramen ovale too large
ASD primum	Near AV valves	Failure of septum primum to fuse with the endocardial cushions
Patent foramen ovale	Common (25% of adults)	Septa form but never permanently fuse
Common atrium	No septation at all	Complete failure of atrial septation

Sinus venosus and the right atrium

The sinus venosus (the receiving chamber, with a **right and left horn**) has horns with **different fates**.

Left sinus horn shrinks -> forms the **coronary sinus** (collects the heart's own venous blood, drains into the RA) and the **oblique vein of the left atrium** (a minor remnant).

Right sinus horn enlarges -> absorbed into the RA wall to form the **smooth-walled sinus venarum** (where SVC and IVC open), and the **valve of the IVC (Eustachian valve)**.

The two parts of the adult right atrium

Part of RA	Origin	Texture
Rough/trabeculated (pectinate muscles)	Original primitive atrium	Rough
Smooth (sinus venarum)	Absorbed right sinus horn	Smooth

Part of RA	Origin	Texture
Boundary	**Crista terminalis** – a muscular ridge (externally: sulcus terminalis)	–

Clinical relevance

- **Persistent left superior vena cava (PLSVC)** – the left anterior cardinal vein fails to regress, leaving a **left SVC that drains into the coronary sinus** (which itself came from the left sinus horn). Usually there is **still a right SVC**, so blood still reaches the RA by a longer route – **most people are asymptomatic**; it is the **commonest thoracic venous anomaly**
- **Coronary sinus ASD (unroofed coronary sinus)** – the coronary sinus is open to the left atrium, creating a shunt

Pulmonary veins & conduction

The **left atrium** gets its smooth wall by a **parallel but different** process – **absorbing the pulmonary veins**. The LA grows an outgrowth toward the lungs, meets the developing pulmonary veins, and **progressively absorbs** them, so **one trunk becomes four pulmonary vein openings** in the adult LA.

Chamber	Source of smooth wall	Mechanism
Right atrium	Right sinus horn	Absorbed into the RA wall
Left atrium	Pulmonary veins	Absorbed into the LA wall

Both atria end up with **smooth posterior walls and rough anterior walls** (the rough part from the original primitive atrium).

Conduction system

Specialised cells in the **right atrium near the sinus venosus** form the **AV node** (receives the atrial signal, delays it before the ventricles) and the **Purkinje fibres** (rapidly conduct through the ventricular walls).

Clinical relevance

- **Anomalous pulmonary venous drainage (APVD)** – the pulmonary vein outgrowth fails to connect to the LA. **Partial** APVD: some veins drain to the wrong place (RA, SVC, IVC). **Total APVD (TAPVD):** all four drain abnormally so **no oxygenated blood enters the LA directly – a surgical emergency in the newborn**
- **Conduction anomalies** – because the conduction system develops near the sinus venosus, defects there (e.g. sinus venosus ASD) can cause **sick sinus syndrome and heart block**

Ventricular septation

Unlike the atria (where the wall grows **down** from the roof), the ventricular septum grows **up from the floor** – and it is **assembled from four components** that fuse:

1. **Inlet** (ventricle)

2. **Outlet** (ventricle)
3. **Trabecular / muscular** (ventricle)
4. **Membranous septum**

   This is **why VSDs are the most common congenital heart defect** – there are **four separate opportunities** for something to go wrong, each producing a VSD in a different location.

The membranous septum

The **last part to close** and the **smallest** – a tiny fibrous patch **just below the aortic valve, adjacent to the tricuspid valve**. Because it closes last, it is the most vulnerable:

> **Perimembranous VSD** – a hole at/near the membranous septum – accounts for ~**80% of all VSDs**.

Ventricular and cono-truncal septation – the developing outlet and membranous septum

Cono-truncal septation

Even after the ventricular septum is complete, the heart has **two ventricles but only one outflow pipe**, made of two stacked segments:

- **Bulbus cordis (conus)** – the lower segment, above the ventricles
- **Truncus arteriosus** – the upper segment, connecting to the aortic arches

This **cono-truncal region must be split lengthwise** into the **aorta (for the LV)** and the **pulmonary trunk (for the RV)**. The dividing wall grows in a **SPIRAL** pattern – the **aorticopulmonary ridges** twist around each other as they fuse.

> **Why spiral?** Because in the adult heart the pulmonary trunk **spirals around** the aorta. If septation were **straight instead of spiral -> Transposition of the Great Arteries (TGA)** – aorta off the RV, pulmonary trunk off the LV.

Neural crest cells – critical

Neural crest cells migrate into the cono-truncal region to populate the ridges; they are **essential for proper fusion**. Without them -> the ridges don't fuse -> **Truncus arteriosus** or **Tetralogy of Fallot**.

The cono-truncal septum also grows down to **close the outlet part of the IVS** – which is why **outlet VSDs are often associated with cono-truncal anomalies**.

Defect	Mechanism
Truncus arteriosus	Complete failure of ridge fusion – a single common arterial trunk persists
Transposition of the Great Arteries (TGA)	Septation without the spiral – straight division puts aorta over RV, pulmonary trunk over LV
Tetralogy of Fallot	Unequal division of the conus – pulmonary side too small -> pulmonary stenosis + overriding aorta + outlet VSD + RVH
Double Outlet RV	Abnormal wedging + neural crest failure – both vessels over the RV

General anatomy of the mature heart

Summary of septation processes

Region	Mechanism	Key clinical failure
Atria	Two septa + foramen ovale valve	ASD, patent foramen ovale
AV canal	Endocardial cushions	AVSD – common in Down syndrome
Ventricles	Four components fusing	VSD – most common CHD
Outflow tract	Spiral aorticopulmonary ridges + neural crest	TGA, Truncus arteriosus, TOF

The aortic arches

The great vessels (aorta, pulmonary arteries, carotids, subclavians) develop from **six paired arterial arches** that form **sequentially** around the pharynx. Think of them as **scaffolding** – you build six pairs, then **demolish most and keep only the parts you need**. They appear and regress in sequence (by the time the 6th forms, the 1st and 2nd have gone).

The framework

- **Arches 1 & 2** – regress early, tiny remnants
- **Arch 3** – the **carotid** system
- **Arch 4** – the **aortic arch** system (asymmetric left vs right)
- **Arch 5** – regresses completely, **nothing**
- **Arch 6** – the **pulmonary** system + **ductus arteriosus**

Complete arch summary

Arch	Left derivative	Right derivative
1st	Maxillary artery (remnant)	Maxillary artery (remnant)
2nd	Hyoid + stapedial (remnant)	Hyoid + stapedial (remnant)
3rd	Common carotid, ECA, proximal ICA	Common carotid, ECA, proximal ICA
4th	**Arch of the aorta**	Proximal right subclavian artery
5th	Nothing	Nothing
6th	Proximal left PA + **ductus arteriosus**	Proximal right PA only

Why is the aortic arch left-sided in normal adults? Because only the **left 4th arch** persists to form it.

The **ductus arteriosus** (from the **left 6th arch**) is the fetal shunt from the pulmonary trunk to the descending aorta. At birth it constricts and becomes the **ligamentum arteriosum**. **Failure to close = Patent Ductus Arteriosus (PDA).**

Clinical relevance of arch anomalies

Defect	Arch	Mechanism
Right-sided aortic arch	4th	Right 4th arch persists instead of left
Aberrant right subclavian artery	4th	Artery crosses behind the oesophagus -> **dysphagia lusoria**
Patent ductus arteriosus	6th	Ductus arteriosus fails to close after birth
Interrupted aortic arch	4th	Gap in the aortic arch

Common cardiac defects and their causes

Common cardiac defects and their embryological causes – part 1

Lesion	Cause
Dextrocardia	Heart tube loops to the left instead of the right – **abnormal looping**
Anomalous pulmonary venous drainage	Failure of pulmonary veins to enter the left atrium
ASD	Failure of **atrial septation**
VSD	Failure of **ventricular septation**
Truncus arteriosus	Failure of fusion of aorticopulmonary ridges; failure of neural crest cell migration
Transposition of great arteries	**Non-spiral** division of the bulbus cordis
Hypoplastic left heart syndrome	Failure of neural crest cell migration

Common cardiac defects and their embryological causes – part 2

Lesion	Cause
Double outlet RV	Abnormal wedging + failure of neural crest cell migration
Double inlet LV	Abnormal wedging + failure of neural crest cell migration
Tetralogy of Fallot	Unequal conal division / failure of neural crest cell migration
Interrupted aortic arch	**4th arch** defect
Persistent ductus arteriosus ± proximal pulmonary atresia	**6th arch** defect

References

Lecture material of Dr Stella Oji; standard embryology texts.